The US Army Research Laboratory is developing helmets with EEG sensors to monitor the brain activity of the user soldiers. The US Air Force is developing helmets, incorporating BCI system to monitor pilot's cognitive workload and stress. Défense Advanced Research Projects Agency (DARPA) has initiated research for development of BCI-enabled prosthetic hands and arms that provide feedback to the user's brain and evoke sensations that the user perceives as coming from his/her own hand. DARPA has also initiated a proof-of-principle project, MOANA (magnetic, optical, and acoustic neural access), to create a dual-function, wireless headset capable of 'reading' and 'writing' brain activity to restore lost sensory function without the need for surgery.

Research is on to allow the paralysed soldiers to perform basic tasks like controlling a wheelchair using the brain signals. The US Air Force has initiated projects to provide the quadriplegics the ability to perform basic operations in an F-35 fighter jet simulator with their thoughts. However, the use of the technology involves operational and institutional risks such as creating new operational vulnerabilities and areas of ethical and legal risks.

BrainNet, a proof-of-principle network

A team of neuroscience and computer science researchers have created a network, called BrainNet, which allows three people to communicate using only their minds and an internet connection. The network uses the characteristic that the brain activity can easily synchronize or align with the external signals. For example, a light flashing at 15Hz causes the brain to emit an electrical signal at the same frequency, and if the frequency is changed to 17Hz the frequency of the brain signal also gets changed to 17Hz.

Architecture of BrainNet:

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In an experiment, the BrainNet connected two 'senders' and one 'receiver,' located in separate rooms, to play a video game in which a falling block had to be rotated to fit into a line at the bottom of the screen. The senders were equipped with EEGs that recorded their brain's electrical activity. The receiver, attached to a TMS, acted as per the received cues from the senders. The two senders selected the option by focussing on the lights flashing on either side of the screen—one at 15Hz and the other at 17Hz.

The brain responses of the senders, captured by the EEGs, were transmitted over the internet to the TMS connected to the receiver. The TMS delivered a magnetic pulse to the receiver's brain causing a transcranial magnetic stimulation, which caused a flash of light (a phosphene) in the receiver's visual field making him act as per the option selected by the sender. Once the receiver acted, the game concluded.

Fig. 8: US Army research in progress (Source: www.factor-tech.com)

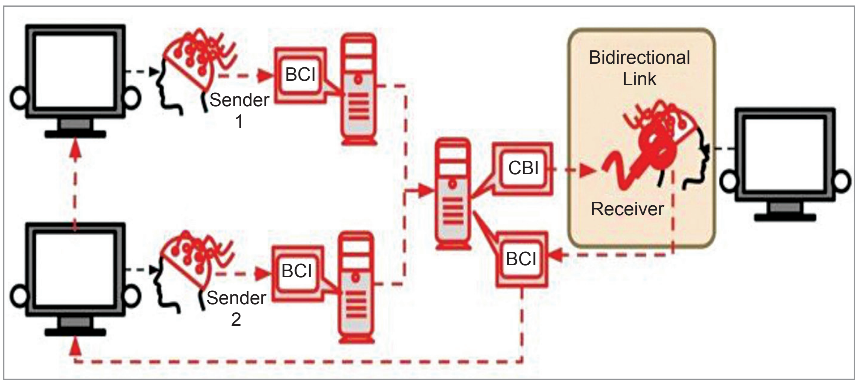

Fig. 9: Architecture of BrainNet (Source: www.nature.com)

Areas of concern

The brain computer interfaces and the associated technologies raise ethical concerns such as managing patient expectations, the concept of personal identity, and the validity of informed consent. One significant ethical issue is getting an informed consent from the patients. It is necessary to be able to distinguish if the consent obtained is fully informed or is affected by the disability. Patients and their relatives are likely to have high expectations from BCI technologies that may not get met. Not working up to the expectations may cause significant distress to them, outweighing the possible benefits.

Use of implantable devices might pose risks for the patient's health. The ability to read minds has a profound impact on the concept of personal identity and privacy. It is important to consider the ethical implications such as safety and security of the data transmitted and stored; breaching the ownership of personal information and data; and precluding other persons or the hackers accessing 'thoughts.'

Areas of concern that need to be explored further include unknown long-term effect; effects of the technology on the quality of life of the subjects and their families; health-related side-effects such as the sleep quality, normal brain functioning and memory; and non-convertibility of the changes made to the brain.

There are also several legal and social issues that need to be settled such as the accountability of and the responsibility for the influence of BCIs; inaccurately translating the cognitive intentions; possible personality chang-